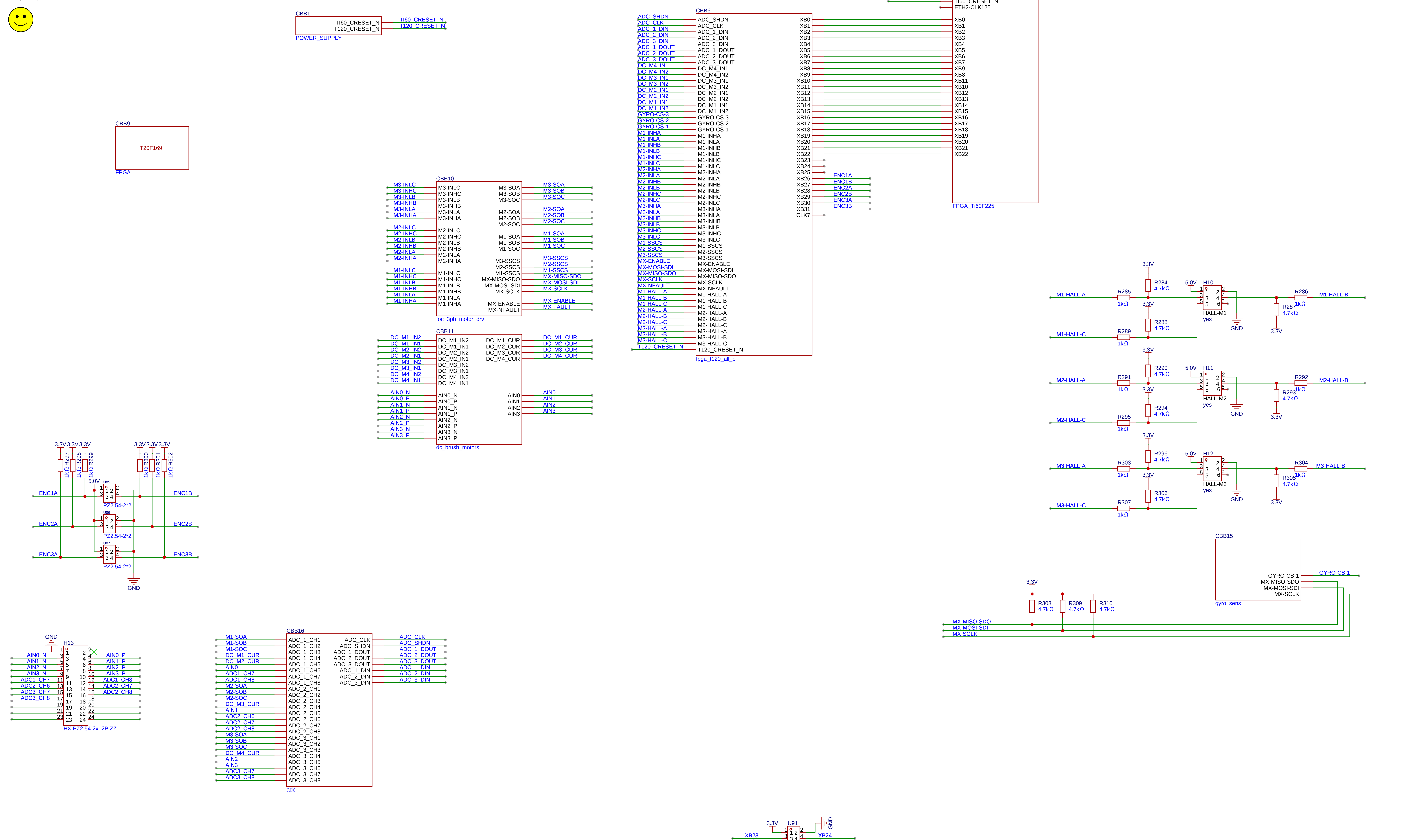
3x3Motor FPGA EcoFlexAI. Delta arm sorting robot

3x3MotorFPGA
18-36V 3x motor controller board with multi camera input
Designed by: Olle Welin 2025

V3.0



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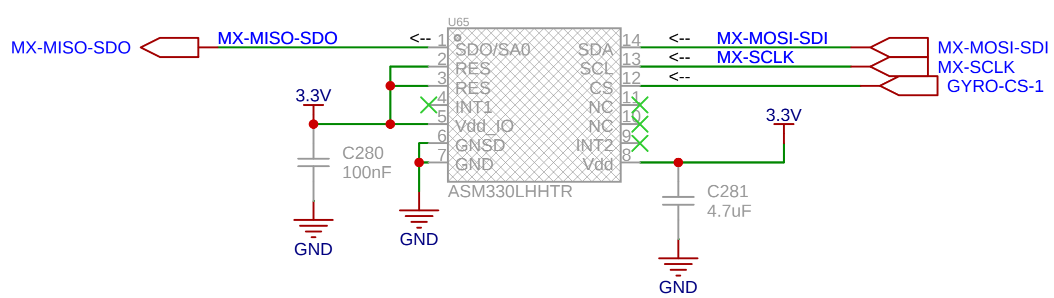
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V3.0	A4	EasyEDA.com

Gyro Sensor 4 On board for cancle out vehical frame movments



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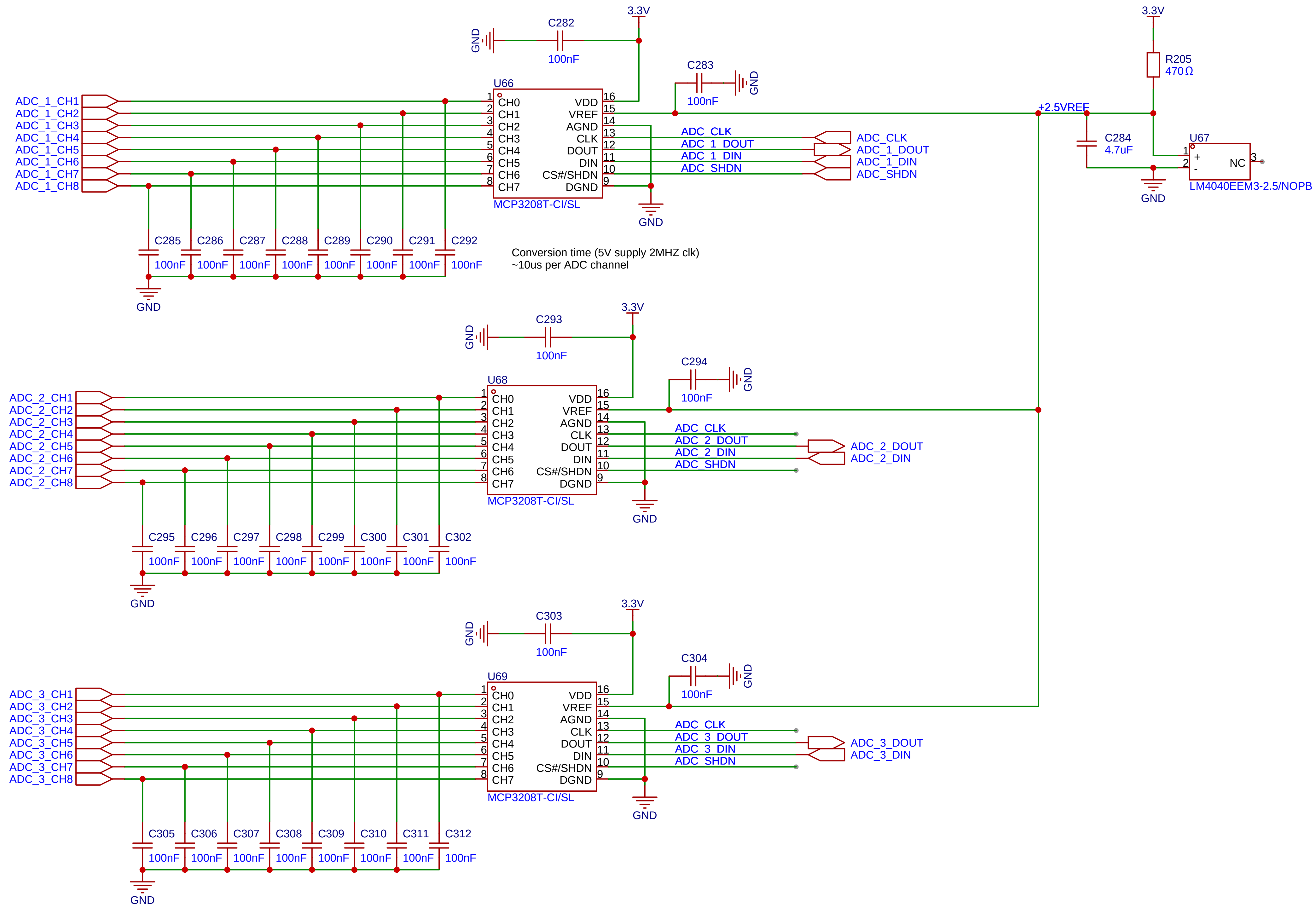

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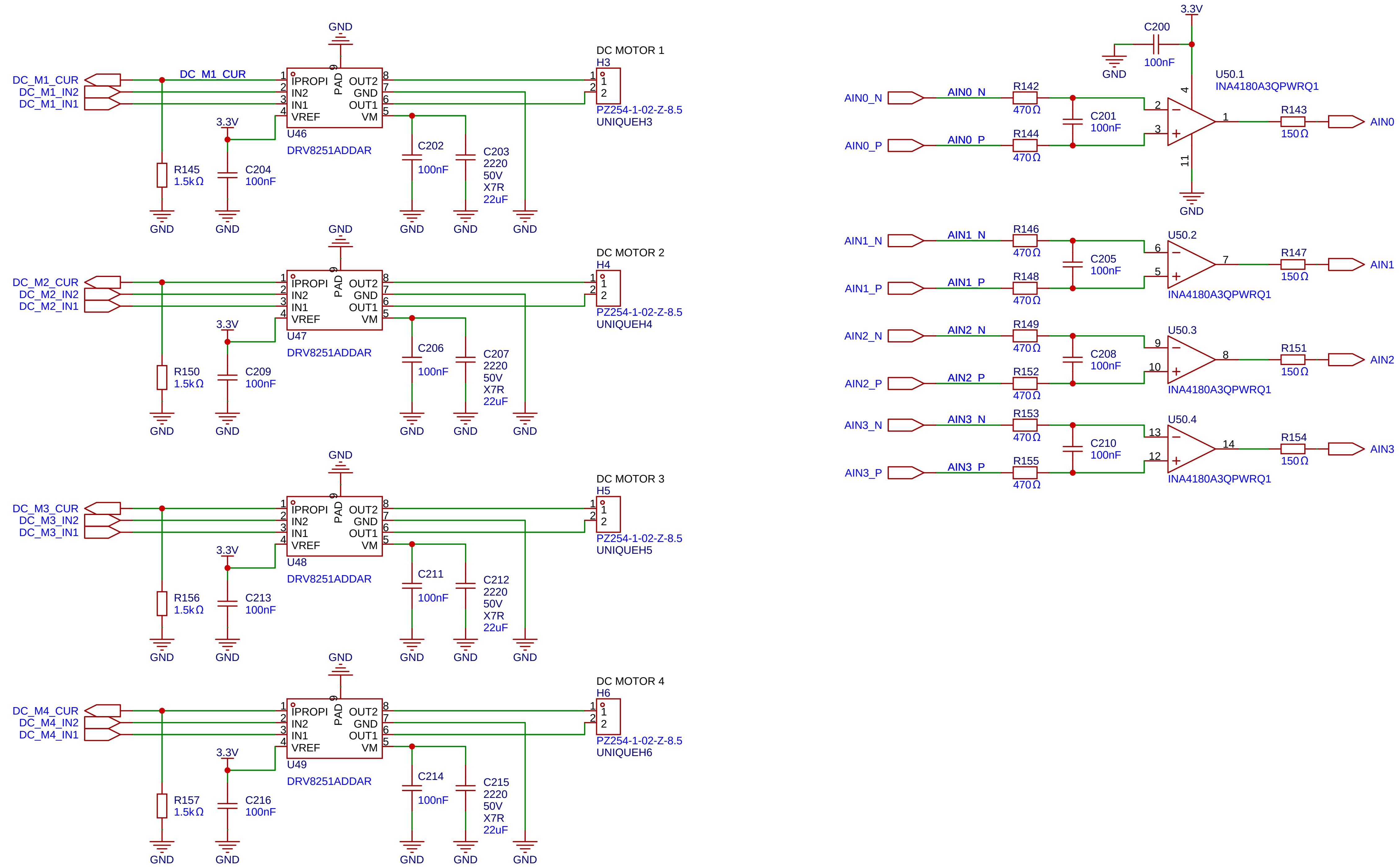

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Reviewed			
		Version	Size
EasyEDA		V1.0	A4

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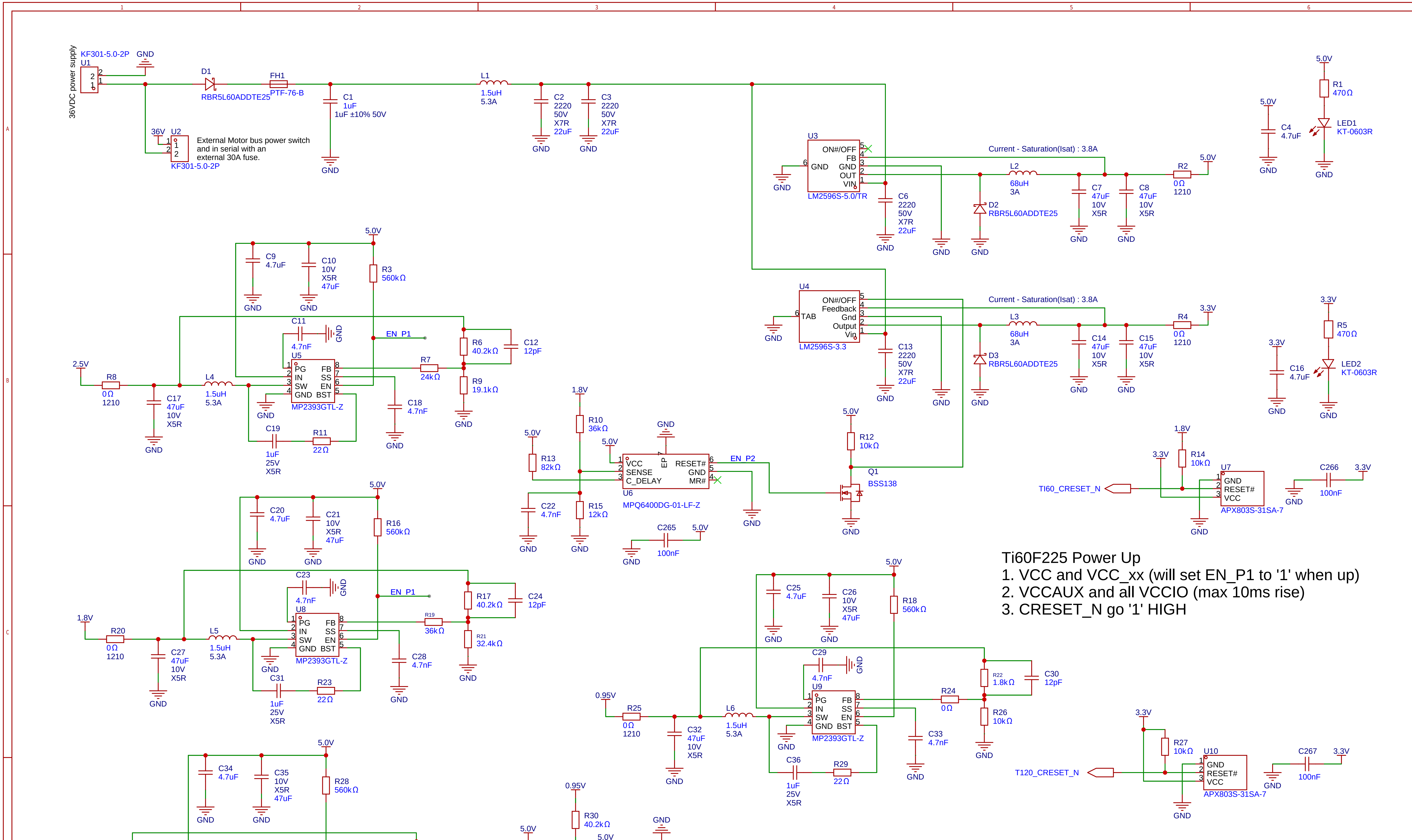
EasyEDA.com



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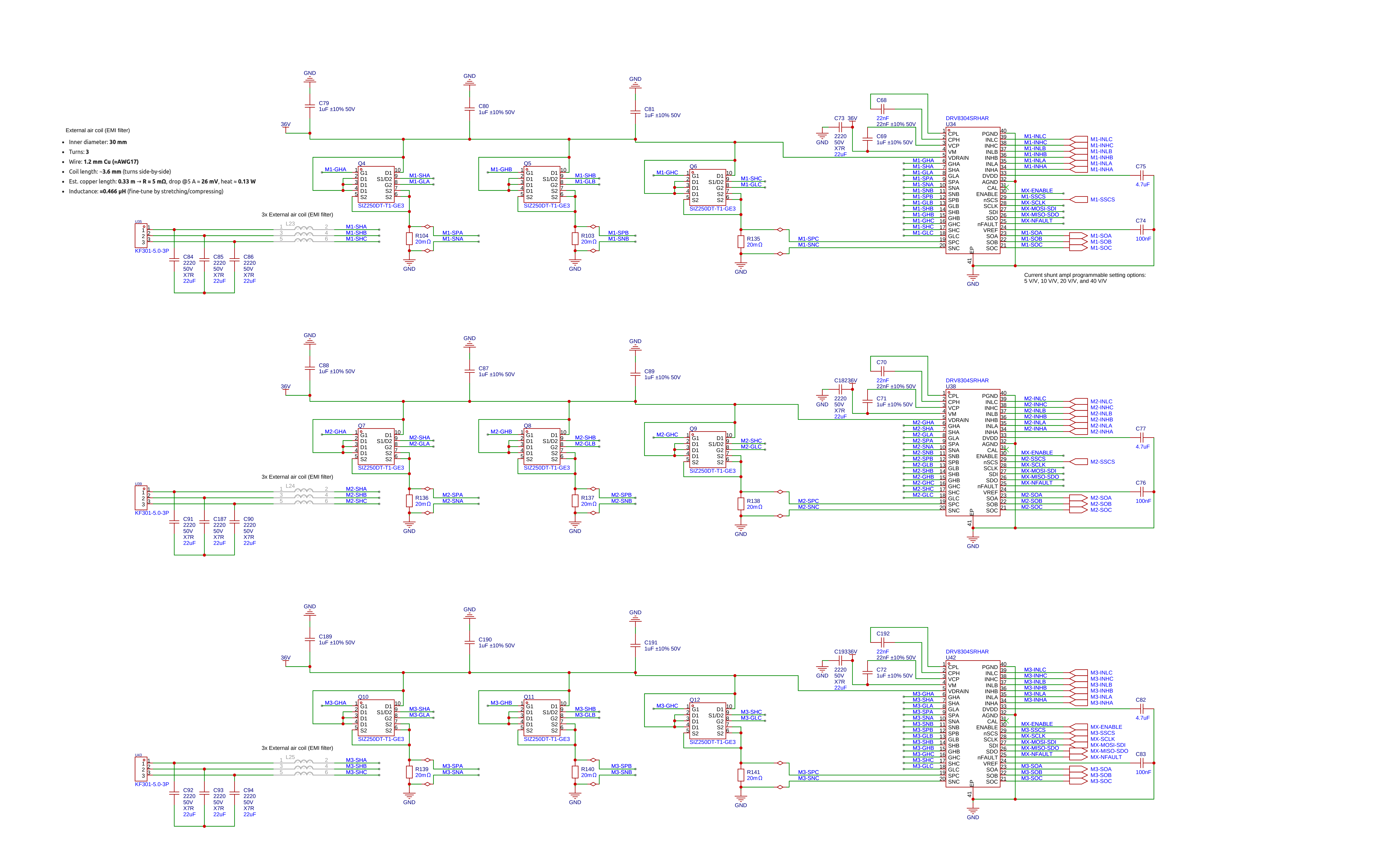
EasyEDA	V1.0	A4	EasyEDA.com



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Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

---------- 17. MIPI RX Lane Usage Summary (begin) ----------

Instance Name	Resource	Pad Names	Package Pins	Clock Region	Group	Mode	Conn Type	LP Reverse	FIFO Crossing	Byte Clock	Delay Mode	Delay
cam_ck[0]	GPIOR_PN_02	GPIOR_N_02_CDI25,GPIOR_P_02_CDI24	M15,M14	R1	I6	clock lane	gclk	Disable	Disable	i_cam_ck_CLKOUT[0]		0
cam_d0[0]	GPIOR_PN_00	GPIOR_N_00_CDI22,GPIOR_P_00_PLLIN0	K11,L11	R0	I6	data lane		Disable	Enable	i_cam_ck_CLKOUT[0]	static	5
cam_d1[0]	GPIOR_PN_01	GPIOR_N_01_CDI23,GPIOR_P_01_EXTFB	L13,L12	R0	I6	data lane		Disable	Enable	i_cam_ck_CLKOUT[0]	static	5

---------- MIPI RX Lane Usage Summary (end) ----------

✅ 7. Summary (What you should do)

✓ Ti60 does NOT have hardened MIPI RX

→ You must use **soft RX** + LVDS-style 1.2 V I/Os.

✓ Use the dedicated "cam_xx" differential pins in Bank 3 (1.2 V)

✓ Two RPi Cameras = 6 differential pairs

You have enough usable pairs.

✓ Recommended pin usage

Camera 0:

- CLK: M14/M15
- D0: L11/K11
- D1: L12/L13

Camera 1:

- CLK: K10/J10
- D0: K12/J12
- D1: L15/K15

✓ Add external termination resistors and LP-mode biasing.

Power Pin Checklist

Check	Pin Name	Package Pins	Description
☐	All	All	I have used the power estimator to ensure that the system power requirements are met.
☐	All	All	My system follows the connection requirements for unused resources and features. See guideline.
☐	All	All	My system follows the power-up sequence. See guideline.
☐	All	All	My system uses the recommended number of decoupling capacitors. See guideline.
☐	VCC	E13 G7 G9 H6 H8 J7 J9 M3	The core power supply voltage is 0.95 V +/- 30 mV or 0.85 V +/- 30 mV.
☐	VCCAUX	D4 E11 L4 M11	The auxiliary power supply is 1.8 V +/- 50 mV.
☐	VCCA_xx	E5 G10 L10 L5	☐ The PLL analog power supply voltage is 0.95 V +/- 30 mV or 0.85 V +/- 30 mV. ☐ The PLL analog power supply has a separate filtering circuit.
☐	VCCIO33	B13 B4 N15 P4	The HVIO bank power supply is 1.8 V, 2.5 V, 3.0 V, or 3.3 V.
☐	VCCIOxx	C11 C7 D2 D6 D9 F14 F4 G12 J11 J4 L14 L2 N11 N5 N7 N9	☐ The HSIO bank power supply to HSIO configured as GPIO is 1.2 V, 1.5 V, or 1.8 V. ☐ The HSIO bank power supply to HSIO configured as LVDS is 1.8 V. ☐ The HSIO bank power supply to HSIO configured as MIPI lane is 1.2 V. ☐ The HSIO bank power supply matches the external configuration interfaces. ☐ The system does not mix different I/O voltages in the same bank.
☐	VQPS_GND	G6	VQPS_GND must always stay low unless you are blowing the security fuses. See guideline.

Schematic	FPGA_Ti60F225	Create at	2025-10-31
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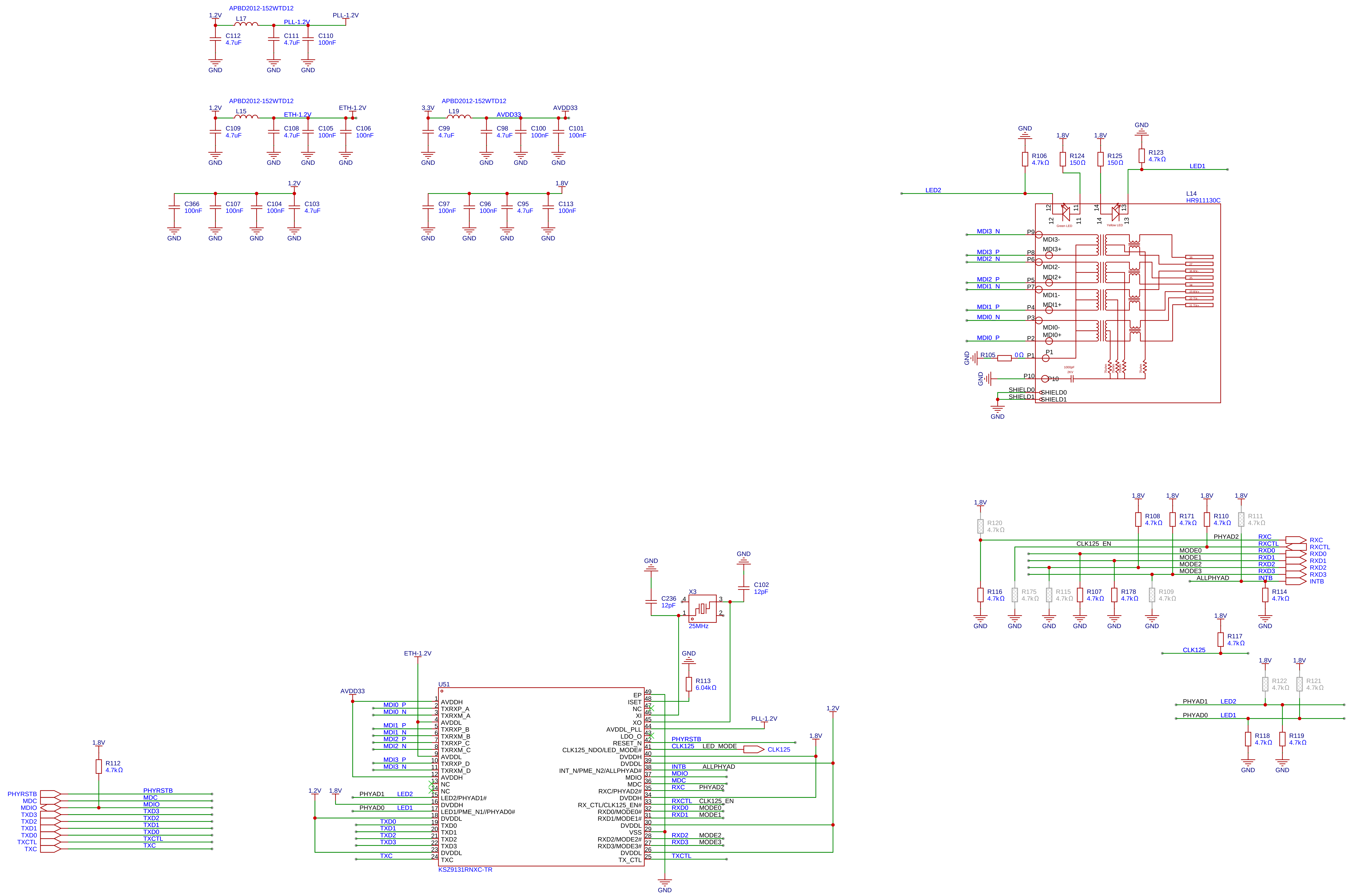

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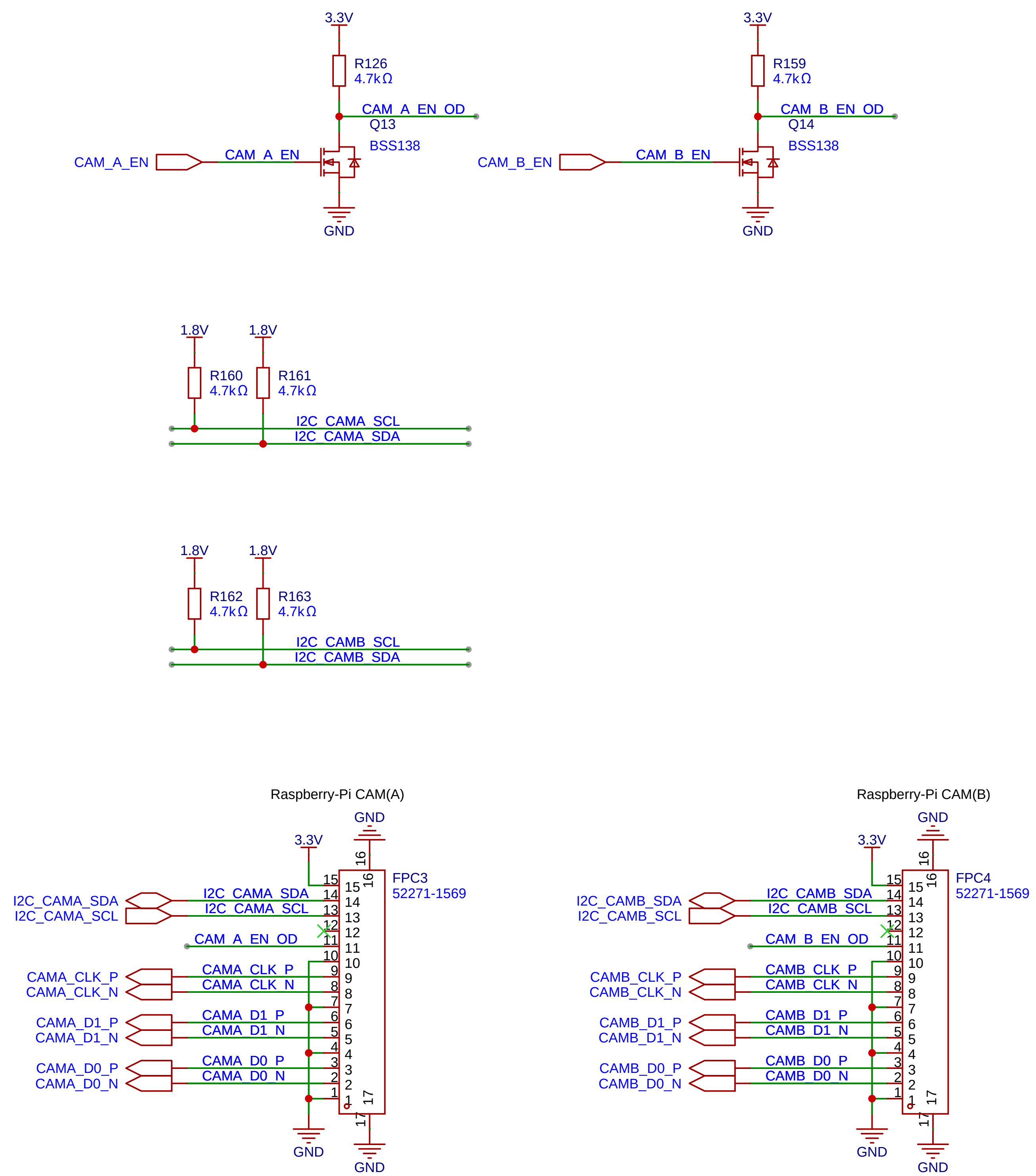
EasyEDA	V1.0	A4	EasyEDA.com



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Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Schematic	fpga_t120_all_p	Create at	2025-11-25
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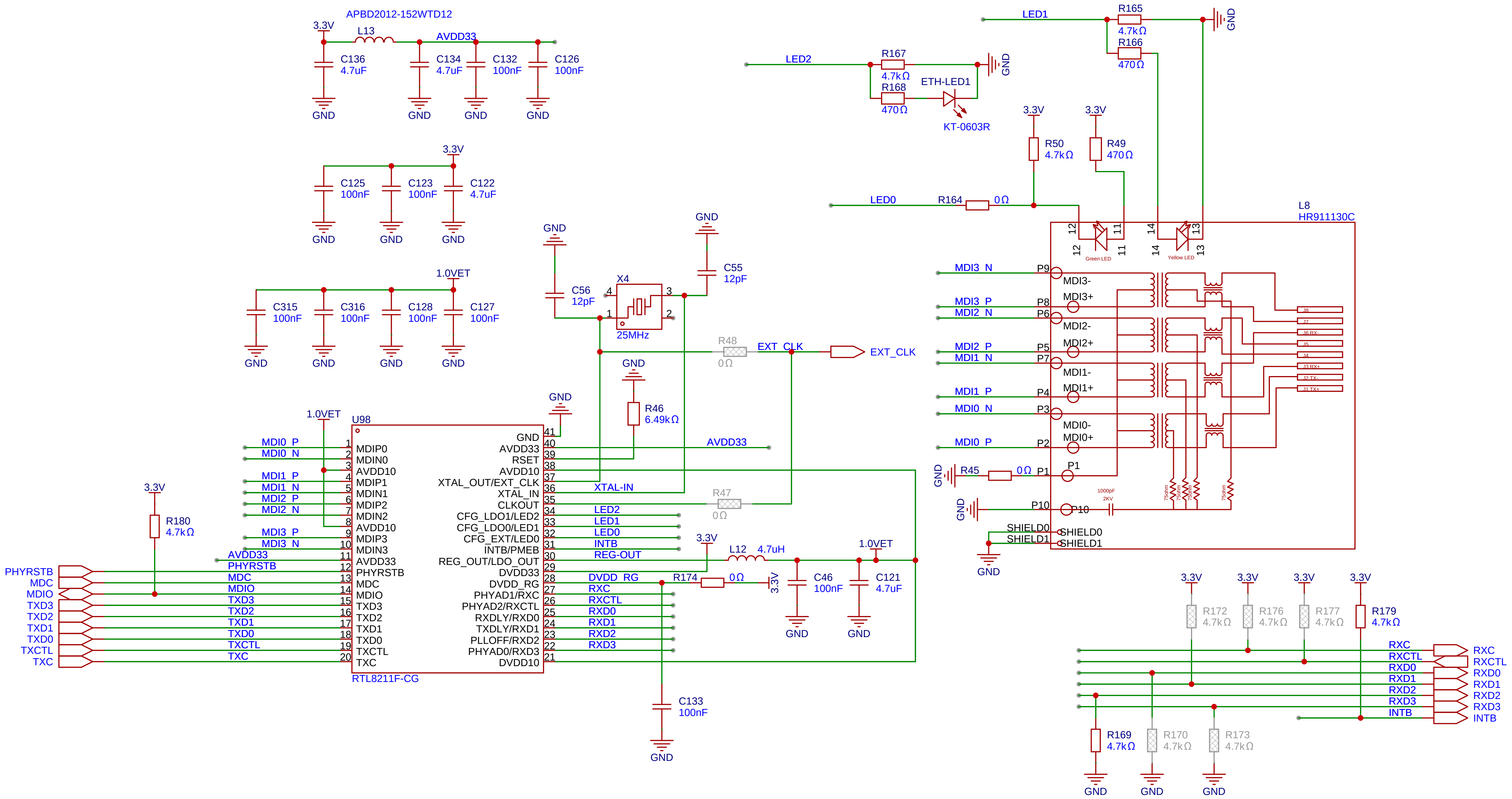
EasyEDA	V1.0	A4	EasyEDA.com



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Version	Size	Page 1 Total 1
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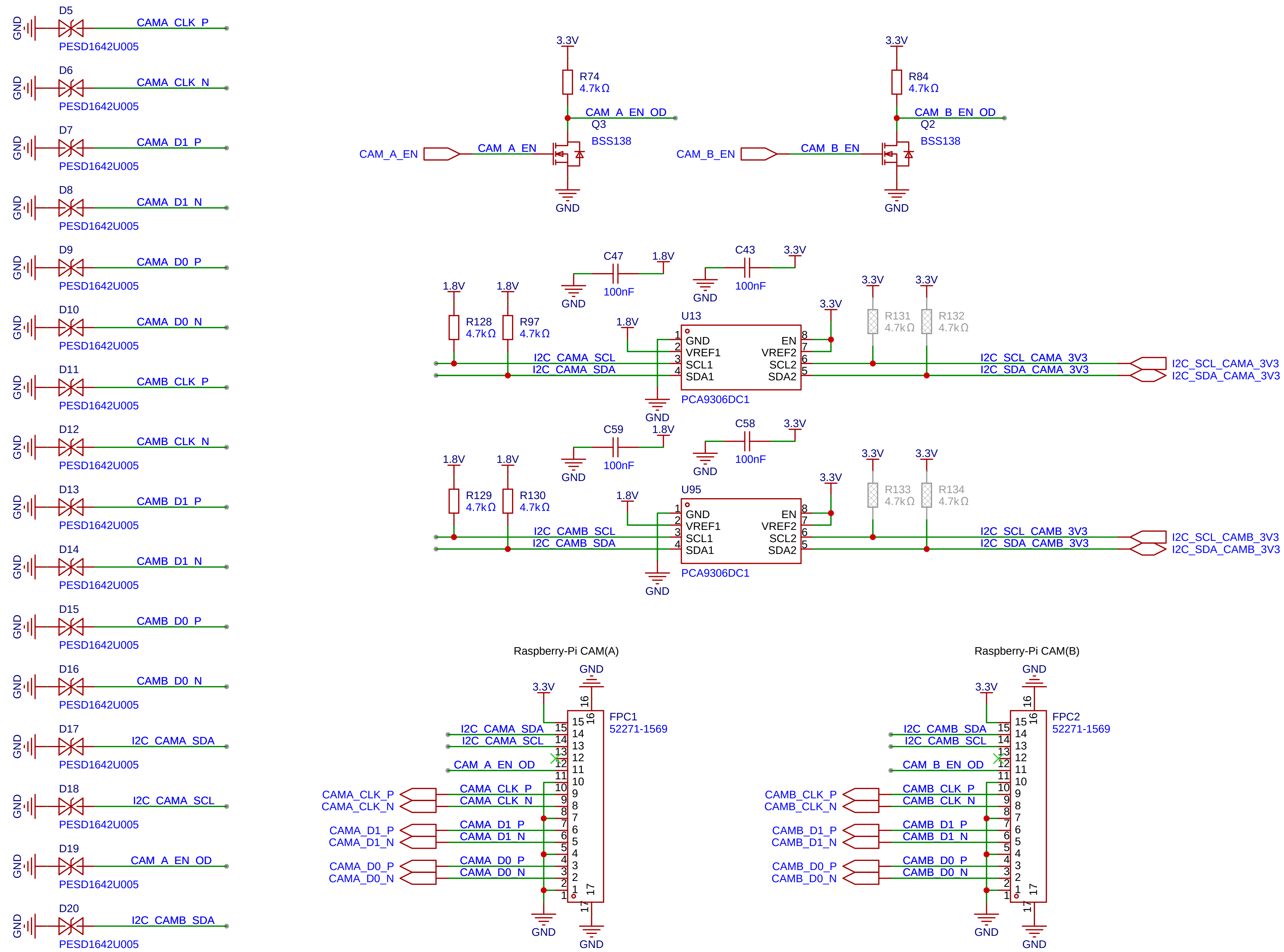
EasyEDA



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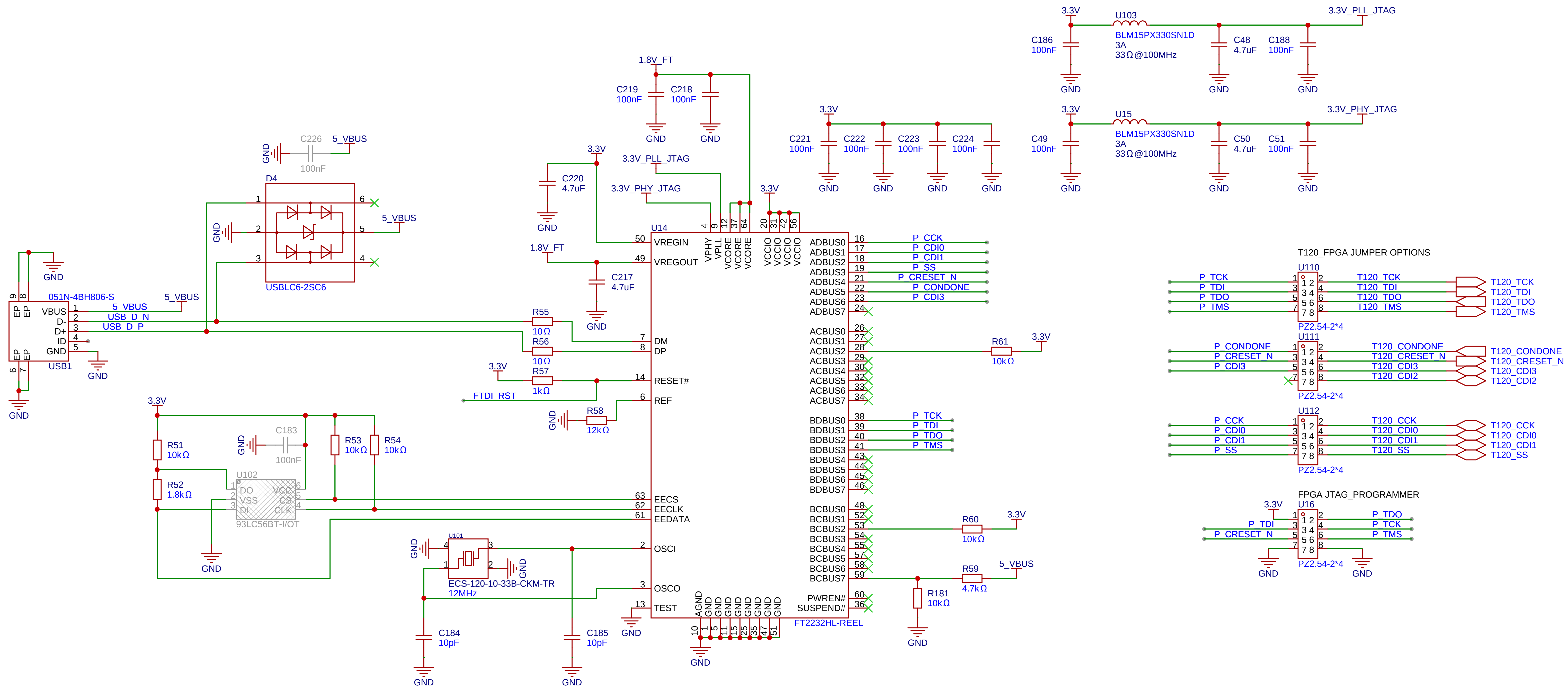

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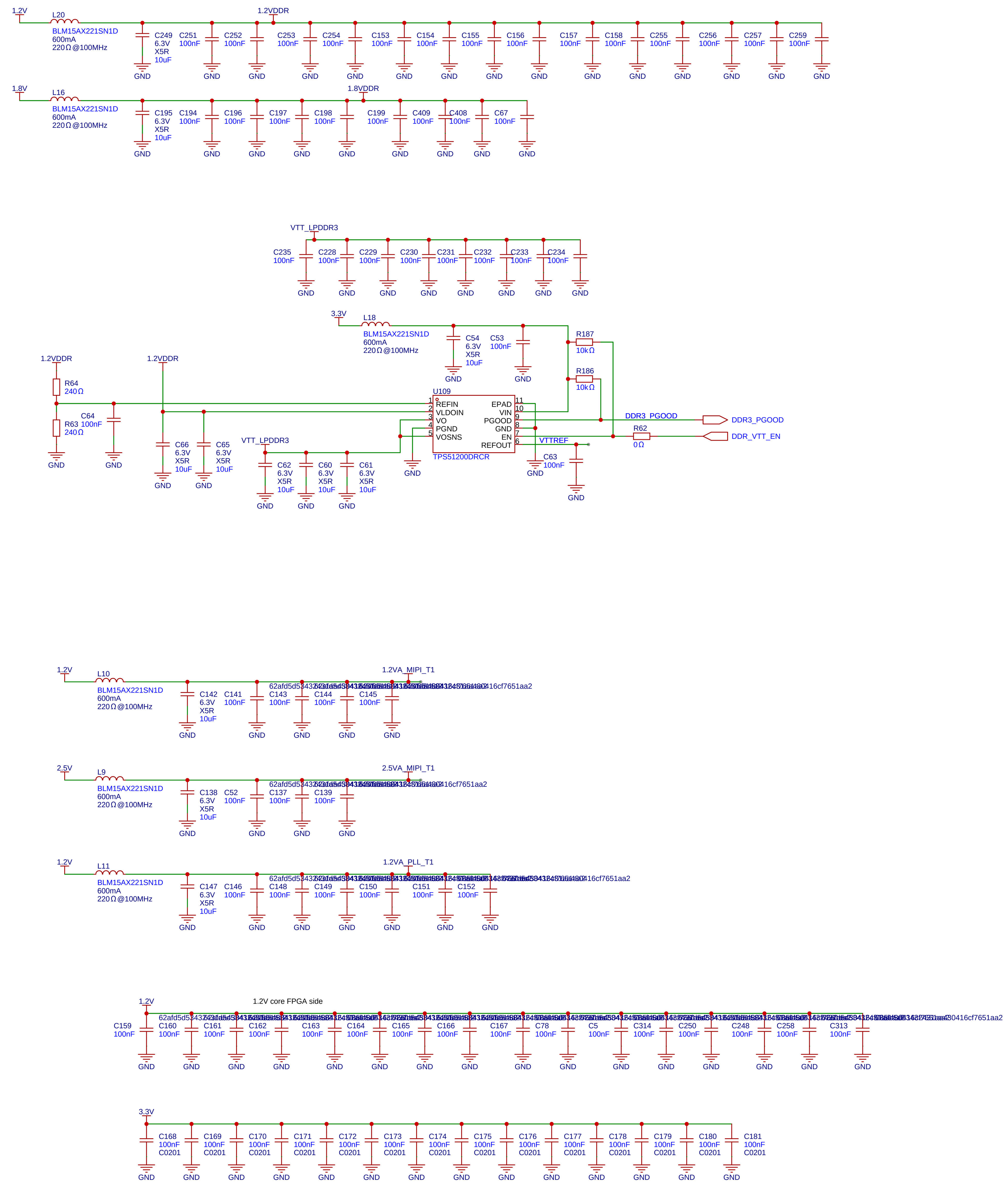

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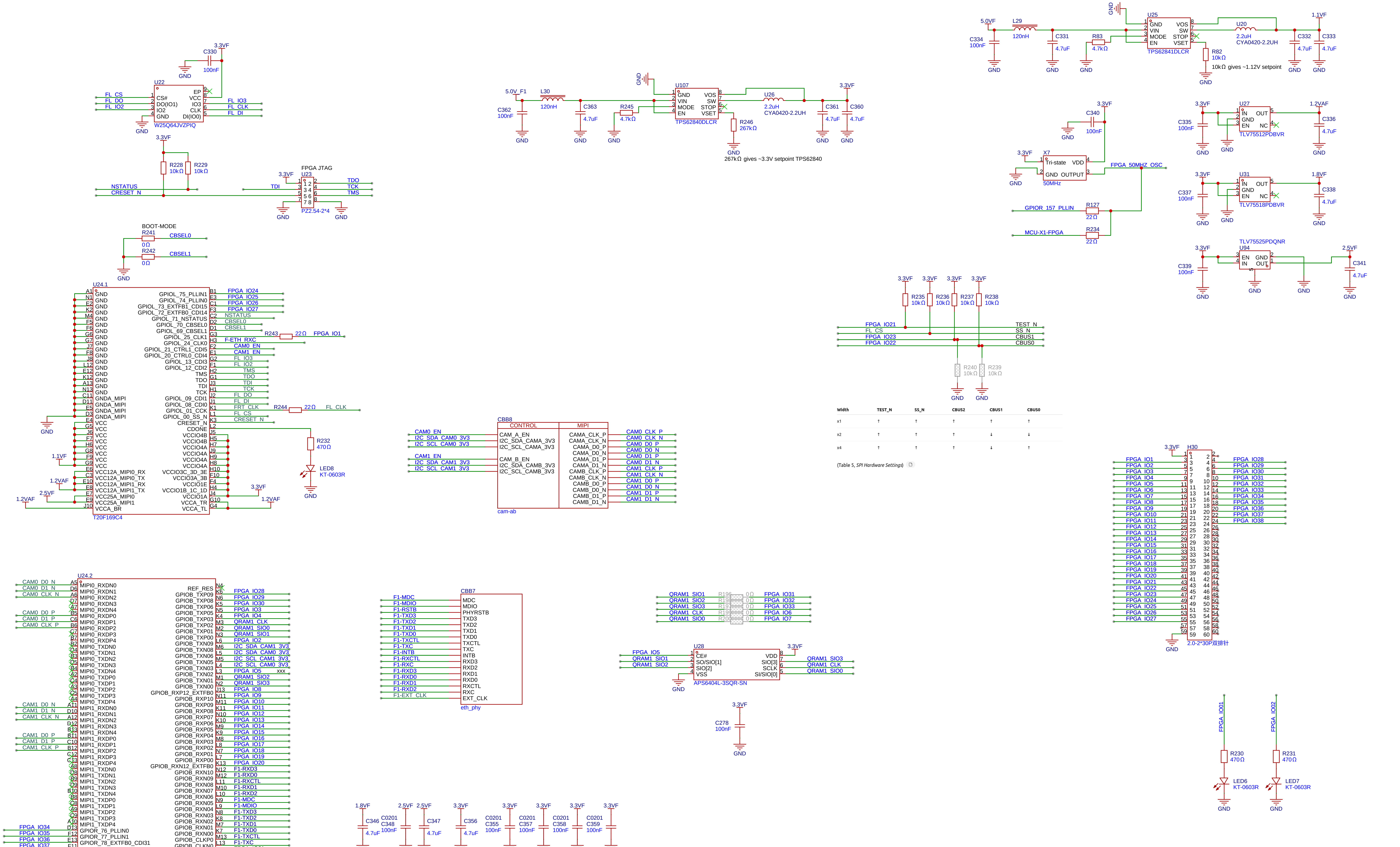

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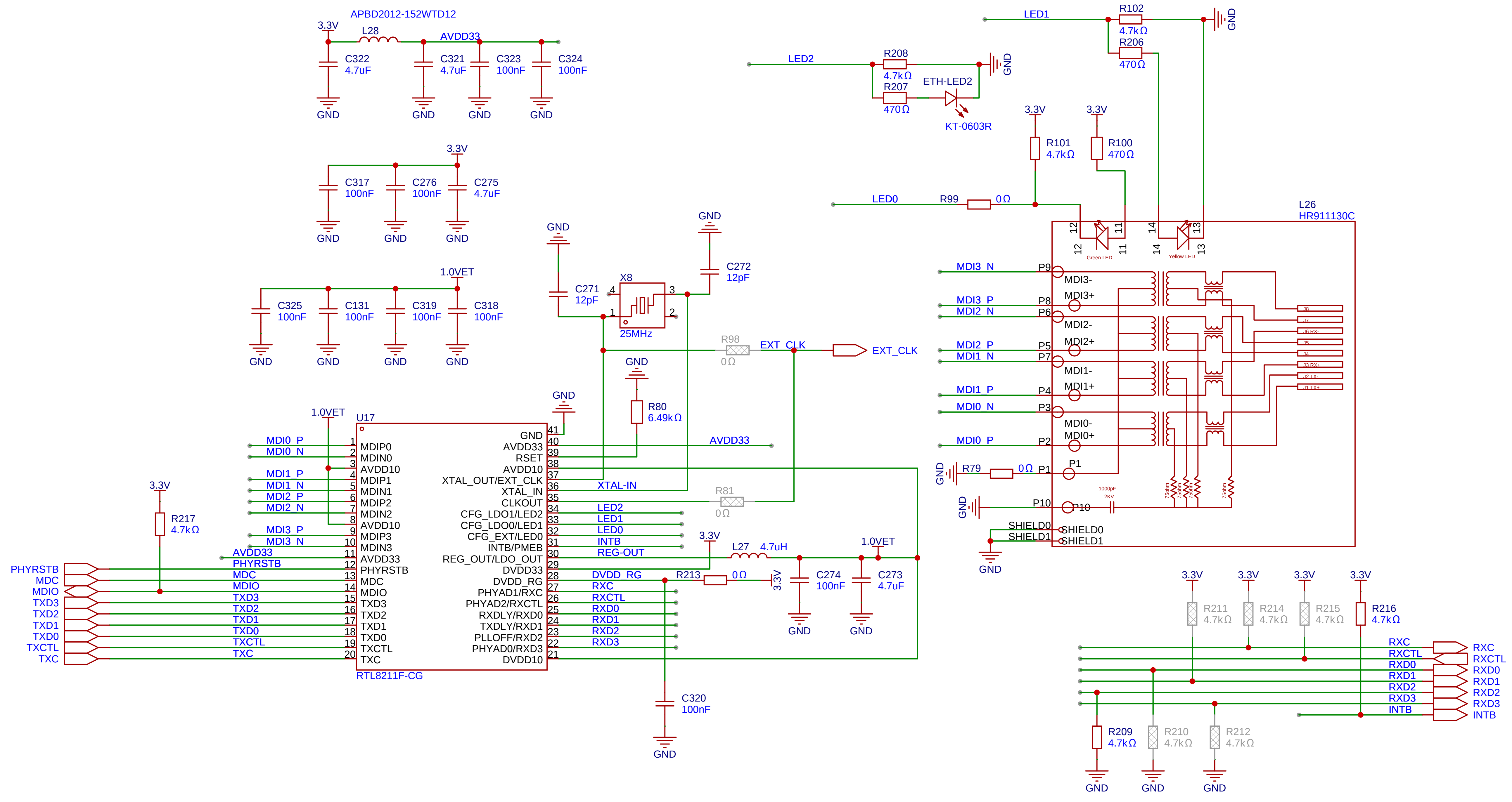

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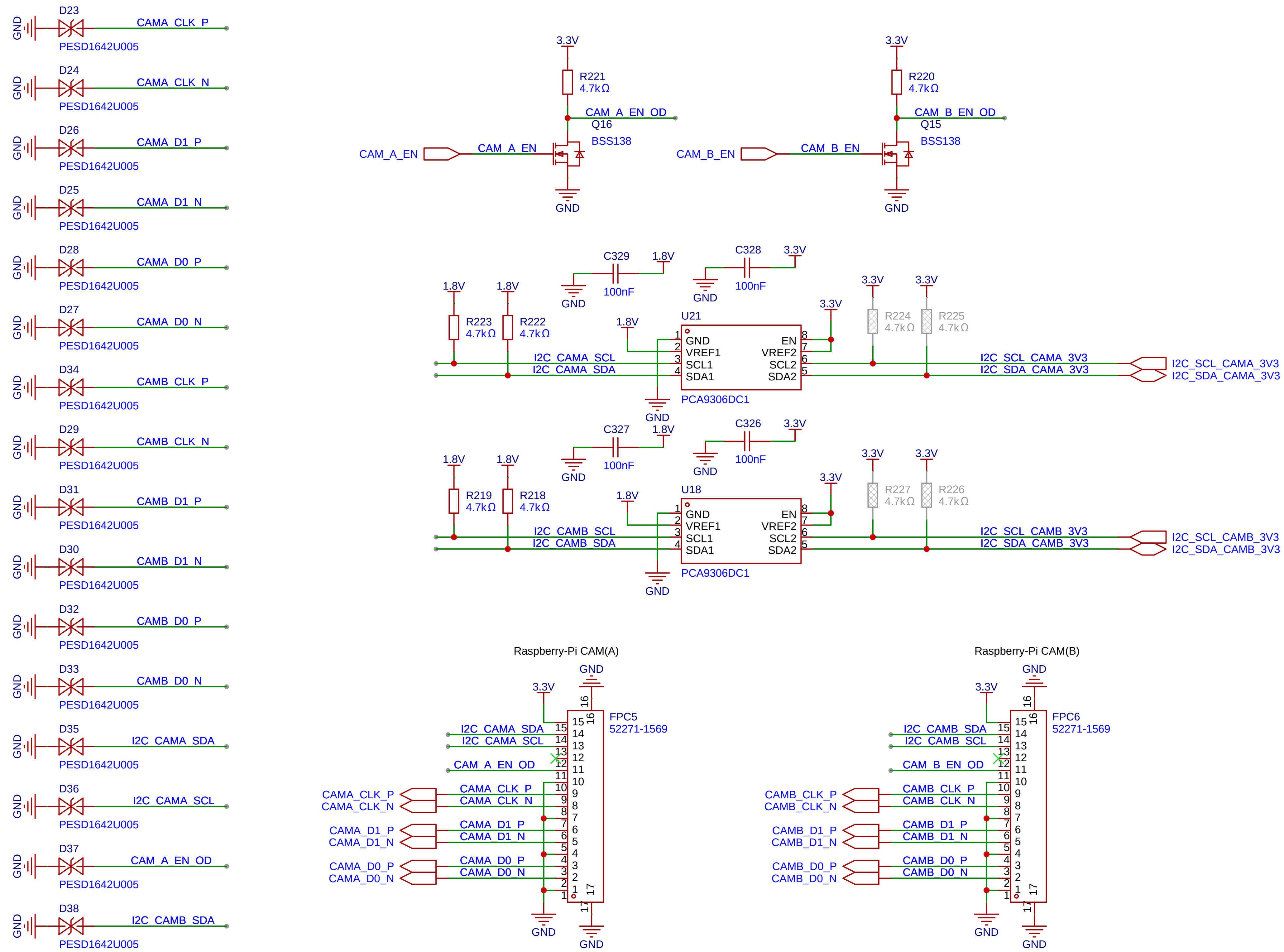

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EasyEDA		V1.0	A4	EasyEDA.com